

GlassBox Options

What It Actually Does — Explained in Plain English

"An AI that suggests trades, but is never the one who's allowed to place them."

The One-Sentence Version

GlassBox Options is a robot that watches the stock options market, has an AI suggest a trade, then makes a completely separate, non-AI "referee" double-check that trade against hard safety rules before anything is ever actually bought or sold — and it keeps a tamper-proof receipt of every decision so a human can verify, after the fact, that it did exactly what it says it did.

Why This Project Exists

AI models are increasingly being pointed at real money — "let the AI trade my portfolio." The problem: AI models can be confidently wrong, can be tricked, and their internal reasoning is hard to audit after the fact. This project is a demonstration of how to make that setup safe: the AI is only ever allowed to *suggest*, never to *act*. A separate, simple, predictable piece of code is the only thing allowed to touch the broker, and it will only do so if the trade passes a fixed checklist of numeric limits (things like "never risk more than 5% of the account on one trade").

The Core Idea: A Wall Between the AI and the Money

Think of it like a restaurant kitchen. The AI is the **sous-chef** who tastes the ingredients and suggests a dish. But the sous-chef never touches the register. A **head chef** (a plain, predictable rulebook — not an AI) has to approve the dish against a fixed list of standards before it goes out. If the sous-chef suggests something with a spoiled ingredient, the head chef rejects it — every time, no exceptions, no matter how convincing the sous-chef's argument was.

In this project, that "head chef" is called the **Risk Kernel**. It is ordinary, deterministic code — the same input always produces the same output. It never calls an AI model. It just checks numbers against limits, like a spreadsheet formula.

How a Trade Actually Happens, Step by Step

1

Look at the market (Perception)

The system pulls real, live data: the stock's current price, how "nervous" the market is about that stock right now (implied volatility), how much it has actually been moving lately (realized volatility), and the real list of tradable options contracts from Alpaca (a real brokerage). This all comes from real data feeds, not guesses.

2

Ask the AI for an idea (Reasoning)

An AI model (Google Gemini) looks at that data and proposes one specific, structured trade — e.g. "sell this option, buy that one, as a hedge, because volatility looks rich compared to how much the stock is actually moving." The AI's answer has to fit a strict template; it cannot just say "buy some stock," it has to specify every leg of the trade precisely.

3

The non-AI referee checks it (Risk Kernel)

Plain code — no AI involved at all — checks the proposed trade against about ten fixed rules: Is the risk capped (no "unlimited loss" trades allowed)? Is it too big a chunk of the account? Would it push the portfolio's directional risk or volatility risk past a hard ceiling? Is the market currently too chaotic to trade (VIX spike)? Is the company about to report earnings (extra risk window)? If any rule fails, the trade is rejected — automatically, instantly, and for a documented numeric reason.

4

Only then, maybe, a real order goes out (Execution)

If — and only if — the Risk Kernel approves, a separate module places the real order with Alpaca's paper-trading (practice money, not real capital) account. This is the only piece of code in the whole system allowed to talk to the broker.

5

A tamper-proof receipt is created (Audit)

The instant a decision is made, the system takes everything that was true at that moment — the market data, the account state, the AI's proposal, the kernel's verdict — and runs it through a cryptographic fingerprint (SHA-256 hash), the same technology used to make sure a downloaded file hasn't been altered. Anyone can later recompute that fingerprint from the stored data; if it still matches, nothing was edited after the fact.

The Three Live Demos, In Plain Terms

The project is built around three short, live demonstrations, each proving a different kind of trustworthiness.

Act 1 — "The system says no when it should"

The Happy Path, and the Guaranteed Rejection

First, a normal trade is proposed and (assuming it's genuinely safe) gets approved and actually placed. Then, on purpose, an oversized/reckless trade is proposed — sized specifically to break the volatility-risk limit — to prove the referee really does say no, with the exact number it broke (e.g. "this would push your risk exposure to -397, but your limit is -250"), not just a vague "error."

Act 2 — "The system catches its own books being tampered with"

TradeTrap Defense

A known real-world danger with AI trading bots is that their internal record of "what positions do I own" can get out of sync with what the broker actually shows — a bug, an attack, or corrupted memory could make the bot believe it owns less risk than it really does. This demo deliberately fakes that corruption (adds a phantom position that only exists in the bot's own memory, never on the real broker) and shows that a background watchdog, running every 30 seconds, compares believed-vs-actual and slams an emergency stop (Kill Switch) the moment they disagree — with no human needing to notice first.

Act 3 — "The AI is reasoning about the market, not memorizing tickers"

The Blindfold Experiment

This checks a subtler kind of honesty: is the AI actually reasoning from the numbers it's given (volatility, price trends), or has it just memorized "SPY is a safe boring stock, always suggest the same thing for it"? The system asks the AI for a trade twice on the exact same real numbers — once with the real ticker name (e.g. "SPY") and once with the name swapped for a meaningless label (e.g. "ASSET_04"). If the AI gives the same structure and the same confidence level either way, that's evidence it's reasoning from the actual numbers, not the brand name.

It Also Learns From Its Own Track Record

Every time a trade is closed, the system records what actually happened (profit or loss) and asks the AI to reflect on it: what worked, what didn't. It then nudges a small set of its own dials in response — for example, after a loss it slightly widens its safety buffers and becomes a bit more selective about which setups it's willing to try next. This is intentionally simple and transparent (a handful of adjustable numbers, not a black-box model being silently retrained) so a human can always see exactly what changed and why.

What's Real, and What's Practice Money

Real	Practice / Simulated
Market prices, options chains, and volatility data are pulled live from Alpaca (a real brokerage's data feed) and the CBOE (the exchange that publishes the VIX "fear index"). Orders are placed on Alpaca's real order-matching system.	All trading happens on Alpaca's paper trading account — practice money, not real capital. If real API keys aren't configured at all, the app falls back to a clearly-labeled made-up price feed just so the screen still works for local development.

No real money is ever placed at risk by this system as it currently exists — "paper trading" means Alpaca simulates the fills against real market prices, but no actual dollars change hands.

Who Would Actually Want This

- **A firm considering letting an AI trade for them**, who wants a concrete example of how to keep a hard, auditable boundary between "AI suggests" and "code executes."
- **A regulator or compliance reviewer**, who wants to see that every decision is independently re-verifiable after the fact, not just "trust us, the logs say it's fine."
- **A developer learning how to build safety rails around an LLM** that has access to a consequential real-world action (here, trading; the same pattern applies to any AI-agent-with-real-power problem).

The Short Version, One More Time

"The AI never touches the broker. It only ever gets to whisper a suggestion to a strict, boring, utterly predictable rulebook — and that rulebook is the only thing with its hands on the trigger. Every decision, approved or rejected, leaves behind a tamper-evident receipt anyone can independently check."

A Quick Glossary

Options / Options Contract	A financial contract that gives the right (not obligation) to buy or sell a stock at a fixed price by a fixed date. This project only trades "defined-risk" combinations of these, meaning the maximum possible loss is capped and known in advance — never an unlimited-loss position.
Implied Volatility (IV)	How much movement the options market is currently pricing in for a stock — a measure of how "nervous" traders are, not a prediction of direction.
Delta / Vega	Two standard risk measurements. Delta = how much a position's value moves if the stock price moves. Vega = how much a position's value moves if market nervousness (IV) changes. The Risk Kernel caps both.
VIX	A well-known public index (published by the CBOE exchange) often called the market's "fear gauge." This project pulls the real, current VIX and refuses to trade if it can't.
Kill Switch	An emergency stop that halts all new trading. Can be triggered automatically (e.g. by the reconciliation watchdog) or manually.
SHA-256 Hash	A cryptographic fingerprint of a piece of data. Changing even one character of the original data produces a completely different fingerprint, which is how tampering gets caught.
Paper Trading	Practice trading against real, live market prices, but with fake/practice money instead of real capital.